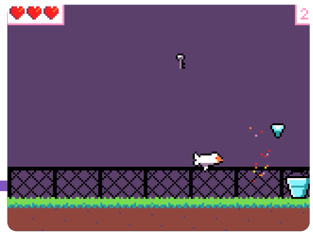
# SWOOP - run sheet

Year 5-6 STEM Launchpad · MakeCode Arcade, Blocks
50 minutes · ~22 students · one laptop each · one teacher



## Before students arrive

- Test the starter and recovery checkpoint on a **school laptop in the student profile**. Confirm Blocks opens, the first edits are visible, and movement works. Preload the starter on every laptop.
- Rehearse **save → reopen → check an actual customisation**. Use the project PNG route below, or a school-approved sharing route. Check network filters, collection permissions and where files land.
- Print A4 at 100%, single-sided. Check one cut set for readability. Prepare mission choices and a STUCK card beside each laptop. Test the trackpad's right-click gesture and classroom volume.
- Try one novice walkthrough: can they find the first edit, restart, make a prediction and recover? Use it to tune the opening pace. A nearby first item now gives an early catching opportunity.

## Agreed 50-minute flow

| TIME | ACTIVITY | TEACHER FOCUS |
|---|---|---|
| **0-3** | Play | “Click the game. Left/right catches shiny things and dodges the crow.” Point to restart. Three lives; SAFE means brief protection after a hit. |
| **3-5** | Welcome + design brief | “You're game designers at Meridan. Make a game someone enjoys.” Show toolbox names, shapes and colours; pan, zoom and undo once. |
| **5-10** | 1 · Make it yours | Choose a sky change or recolour a few bucket pixels. Test visibility and catching. Offer further design work straight away. |
| **10-18** | 2 · Find your fun | Predict, change one number, restart, test 20 seconds. Relaxed, busy and tricky are all valid choices. |
| **18-25** | 3 · Give it a win | Demonstrate the separate on-score event and WIN inside it. Test at 2, then set the chosen target. Offer the working checkpoint for recovery. |
| **25-37** | Choose a challenge | More crows, art, sound, timer or an invention. Students may continue their current change. Help is available to everyone. |
| **37-43** | Partner test + refine | A neighbour plays: “I noticed…” and “What if…?” Give one observation and one suggestion; adjust one thing. |
| **43-47** | Keep the game | Follow the rehearsed save/collection route. Preserve each student's customised version and explain how to reopen it. |
| **47-50** | Celebrate + look ahead | Show 2-3 varied successes: art, an experiment, a feature. “What would you build next?” Link designing, coding and testing to future STEM challenges at Meridan. |

## What success looks like

**One owned change, one prediction/test, and one thing each student can show or explain.** Missions 1-3 provide a supported route to a winnable game; card counts are not completion guarantees. Celebrate a useful discovery or helpful playtest as well as a new feature.

**Protect the closing three minutes.** Show Curiosity through predictions, Creativity through choices, Courage through trying again, and Collaboration through feedback.

# SWOOP - desk reference

## Mission guidance and recovery

| MISSION | WHAT TO NOTICE OR EXPLAIN |
|---|---|
| **1 · Make it yours** | Sky and bucket sit below the two tuning numbers. Recolouring is a small first success. Keep solid pixels: an erased catcher cannot catch. Shape and size can change collisions. Test contrast. |
| **2 · Find your fun** | Smaller `dropEvery` means more frequent loot; larger `fallSpeed` means faster falling. Try 700 or 60 separately. Predict before testing; there is no single correct difficulty. |
| **3 · Give it a win** | The pink `on score` event stands alone. The purple WIN action belongs inside. Test with 2 first; restore the chosen target, such as 15. A partner can help tune a 30-60 second win. |
| **4 · Change the challenge** | Duplicate the whole 3500 ms crow event, then try 2000 ms. This adds a stream of crows. Tune fairness as well as difficulty. Show the actual laptop's right-click gesture. |
| **5 · Your twist** | Choose one. Scale: select `player`, 1.5, below its creation in on start. Sound: existing Player-Shiny overlap, in background. Timer: on start, 60 seconds; zero loses. Fifth item: + at bottom of loot array. |

## When something surprises them

**“Nothing happened.”** Restart after editing. Check actions are inside the intended event. Rounded-top events stand separately.

**“My catcher disappeared.”** Undo the last change (Ctrl + Z; Cmd + Z on Mac), or open its picture and draw solid pixels. If needed, save their work before opening a recovery checkpoint.

**“Too much loot!”** Zero wait creates many sprites. Restore a positive interval, such as 700 or 2500. Earlier desktop stress checks do not promise school-laptop performance.

## Support a deeper challenge

**Reverse crow (advanced):** copy the frame list into a new variable, flip BOTH images horizontally, and select that new list in the copied animation. In the copied event, set x to 166 and vx to -72. Editing the original shared list changes both streams.

**One-hit shield:** plan the trigger, a variable that remembers protection, and visible feedback. Hint ladder: event → state → action. Test one crow and then another to prove it was used.

**Starter protection:** a crow hit starts 1 second of SAFE text and particles. Further crows cannot remove life during that second; catching still works.

## Keep each student's own game

1. Give the project a recognisable name. Click the **Save icon beside the name** to download its project PNG. Keep the original file; an ordinary screenshot is not a project.
2. In a fresh browser/profile, open MakeCode Arcade → **Import** → **Import File**. Select the PNG, open it, then check Blocks, a customisation and play.
3. Collect via the school's agreed route and explain how the student will get their file. If sharing is approved and works, a share link is an alternative; share again after later edits. Do not assume email access.

**Still needs the school rehearsal:** student profile/network, save and collection, actual printing, audio and novice timing. No public starter share URL is available. Use the supplied local starter/checkpoint project files for preloading.